

Recommended Books for BS Artificial Intelligence

Serial	Subject Name	Recommended Books
1	Programming Fundamentals	1. <i>C How to Program</i> by Paul Deitel & Harvey Deitel 2. <i>Starting out with Programming Logic & Design</i> by Tony Gaddis
2	Object Oriented Programming	1. <i>C++ How to Program</i> by Deitel & Deitel 2. <i>Object Oriented Programming in C++</i> by Robert Lafore
3	Data Structures	1. <i>Data Structures and Algorithm Analysis in C++</i> by Mark Allen Weiss 2. <i>Data Structures and Algorithms in C++</i> by Adam Drozdek
4	Computer Org. & Assembly Language	1. <i>Computer System Architecture</i> by M. Morris Mano 2. <i>Computer Systems: A Programmer's Perspective</i> by Randal E. Bryant
5	Digital Logic Design	1. <i>Digital Fundamentals</i> by Floyd 2. <i>Fundamental of Digital Logic with Verilog Design</i> by Stephen Brown
6	Operating Systems	1. <i>Operating Systems Concepts</i> by Abraham Silberschatz 2. <i>Modern Operating Systems</i> by Andrew S. Tanenbaum
7	Database Systems	1. <i>Database System Concepts</i> by Avi Silberschatz, Henry F. Korth 2. <i>Database Management Systems</i> by Raghu Ramakrishnan
8	Information Security	1. <i>Computer Security: Principles and Practice</i> by William Stallings 2. <i>Principles of Information Security</i> by M. Whitman & H. Mattord

9	Computer Networks	1. <i>Computer Networking: A Top-Down Approach</i> by James F. Kurose & Keith W. Ross 2. <i>Computer Networks</i> by Andrew S. Tanenbaum
10	Software Engineering	1. <i>Software Engineering</i> by Ian Sommerville 2. <i>Software Engineering: A Practitioner's Approach</i> by Roger S. Pressman
11	Analysis of Algorithms	1. <i>Introduction to Algorithms</i> by Thomas H. Cormen et al. 2. <i>Algorithms</i> by Robert Sedgewick & Kevin Wayne
12	Artificial Intelligence	1. <i>Artificial Intelligence: A Modern Approach</i> by Stuart Russell & Peter Norvig 2. <i>AI Algorithms, Data Structures, and Idioms</i> by George F. Luger
13	Parallel & Distributed Computing	1. <i>Distributed Systems: Principles and Paradigms</i> by Andrew S. Tanenbaum 2. <i>Distributed and Cloud Computing</i> by K. Hwang & J. Dongarra
14	Calculus & Analytic Geometry	1. <i>Calculus and Analytic Geometry</i> by George B. Thomas 2. <i>Calculus</i> by James Stewart
15	Linear Algebra	1. <i>Elementary Linear Algebra</i> by Howard Anton 2. <i>Linear Algebra and its Applications</i> by Gilbert Strang
16	Probability & Statistics	1. <i>Probability and Statistics for Engineers and Scientists</i> by Ronald E. Walpole 2. <i>Schaum's Outline of Probability and Statistics</i> by John Schiller
17	Functional English	1. <i>College Writing Skills with Readings</i> by John Langan

		2. <i>A Textbook of English Prose and Structure</i> by Arif Khattak
18	Expository Writing	1. <i>Effective Communication Skills</i> by John Nielsen 2. <i>Practical Business English</i> by Collen Vawdrey
19	Technical & Business Writing	1. <i>Technical Report Writing</i> by Pauley and Riordan 2. <i>Effective Technical Communication</i> by Ashraf Rizvi
20	Islamic Studies	1. <i>Introduction to Islam</i> by Dr. Hamidullah 2. <i>Principles of Islamic Jurisprudence</i> by Ahmad Hassan
21	Ideology & Constitution of Pakistan	1. <i>The Emergence of Pakistan</i> by Chaudary M. 2. <i>A Short History of Pakistan</i> by I. H. Qureshi
22	Professional Practices	1. <i>A Gift of Fire: Social, Legal, and Ethical Issues</i> by Sara Baase 2. <i>Computer Ethics</i> by Deborah G. Johnson
23	Discrete Structures	1. <i>Discrete Mathematics and Its Applications</i> by Kenneth H. Rosen 2. <i>Discrete Mathematics with Applications</i> by Susanna S. Epp
24	Applied Physics	1. <i>Fundamentals of Physics (Extended)</i> by Resnick and Walker 2. <i>Physics for Computer Science Students</i> by Narciso Garcia
25	Introduction to ICT	1. <i>Understanding Computers: Today and Tomorrow</i> by Charles S. Parker 2. <i>Computer Fundamentals</i> by Pradeep K. Sinha
26	ANN & Deep Learning	1. <i>Deep Learning</i> by Ian Goodfellow, Yoshua Bengio 2. <i>Neural Network Design</i> by Martin T. Hagan

27	Computer Vision	1. <i>Computer Vision: Algorithms and Applications</i> by Richard Szeliski 2. <i>Multiple View Geometry in Computer Vision</i> by Richard Hartley
28	Knowledge Rep. & Reasoning	1. <i>Knowledge Representation and Reasoning</i> by Ronald Brachman 2. <i>Artificial Intelligence: Foundations of Computational Agents</i> by David Poole
29	Machine Learning	1. <i>Machine Learning: A Probabilistic Perspective</i> by Kevin P. Murphy 2. <i>Machine Learning</i> by Tom M. Mitchell
30	Natural Language Processing	1. <i>Speech and Language Processing</i> by Daniel Jurafsky & James H. Martin 2. <i>Foundations of Statistical Natural Language Processing</i> by Manning & Schütze
31	Programming for AI	1. <i>Python for Data Analysis</i> by Wes McKinney 2. <i>Python for Everybody: Exploring Data Using Python 3</i> by Charles R. Severance
32	Deep Learning (Elective)	1. <i>Deep Learning with Python</i> by François Chollet 2. <i>Deep Learning</i> by Ian Goodfellow
33	Theory of Automata	1. <i>An Introduction to Formal Languages and Automata</i> by Peter Linz 2. <i>Introduction to Computer Theory</i> by Daniel I. A. Cohen

Total number of unique recommended books in this curated list: 63